AN EXPLICIT BOUND OF TWELVE FOR CYCLE–EDGE DECOMPOSITIONS

JIACHENG SHI

Abstract. For a finite simple graph G, let $c(G)$ denote its number of connected components, including isolated vertices. We prove that the edges of G admit a partition into at most $12(|V(G)|-c(G))$ simple cycles and single edges. The argument establishes an explicit weighted bound with cycle weight 1 and single-edge weight 23/42, using a finite-degree completion theorem, three successive rates in a scale decomposition, and a renewal potential. A finite one-vertex amplification and a minimum-weight packing convert the weighted estimate into the stated component-rank bound. Rational certificates supply the quantitative interfaces, including all stopping phases and the infinite tails. A complete Lean formalization of the final graph-theoretic statement accompanies the paper; the retained project source closure was rebuilt and its proof dependencies audited.

Contents

1. Introduction 2
2. Weighted packings and component-rank normalization 3
3. Classical inputs and their formal proofs 4
4. Parameters and the renewal potential 6
5. Finite-degree completion 7
5.1. The short-circumference starting point 8
5.2. The mixed cover 8
5.3. A certified depth-first-search step 9
6. An exact decomposition into scales 10
7. Routing, completion, and original-edge ownership 12
7.1. Two edge banks and smaller-order completion 12
7.2. A single routing system for all requests 13
8. Folding a layer and paying its cost 14
9. The no-fold branch and the infinite tail 16
10. Completion of the proof 17
11. Formal statement and verification artifact 17
Appendix A. The short-circumference reduction and catalogue 18
Appendix B. Certification of the mixed-cover construction 20
B.1. Coordinates and fixed-size samples 20
B.2. From the sample to cycles 21
B.3. Continuous coverage and its soundness interface 21
Appendix C. Exact parameters and certificate indices 21
Appendix D. Correspondence with Lean 26
References 27

Date: November 6, 2026.
2020 *Mathematics Subject Classification.* 05C38, 05C70; 68V15.
Key words and phrases. cycle decomposition, Erdős–Gallai conjecture, weighted graph decomposition, computer-assisted proof, Lean.
Copyright 2026 Jiacheng Shi. This manuscript is licensed under CC BY 4.0.

Acknowledgements and use of AI tools 28

1. Introduction

All graphs in this paper are finite, simple, and undirected. A *cycle–edge decomposition* is a partition of the original edge set into edge sets of simple cycles and singleton edges. Let $\mathrm{ce}(G)$ be the minimum number of parts in such a partition. We write

$$n = |V(G)|, \qquad m = |E(G)|, \qquad r(G) = n - c(G),$$

where $c(G)$ counts all connected components, including isolated vertices. The empty graph on no vertices has $c(G) = 0$.

Theorem 1.1. *Every finite simple graph G satisfies*

$$\mathrm{ce}(G) \le 12\big(|V(G)| - c(G)\big).$$

The problem of obtaining a linear bound for $\mathrm{ce}(G)$ goes back to Erdős and Gallai; it appears in the discussion of Erdős, Goodman and Pósa [7]. Conlon, Fox and Sudakov [4] proved an $O(n \log\log n)$ bound, and Bucić and Montgomery [2] subsequently obtained $O(n \log^* n)$. Linear bounds were presented in 2026 by OpenAI [14], by Kim [12], and in a separate manuscript and formalization of Coffey [3]. The present paper addresses an explicit coefficient. Its large-scale construction follows the quotient-and-lifting approach of [14]; the result is not a new proof of the mere existence of some linear constant.

Our proof first assigns weight 1 to a cycle and weight $\sigma = 23/42$ to a singleton edge. With $C = 485/42$, the central estimate is

$$\exists \mathcal{P} \quad \mathrm{wt}_\sigma(\mathcal{P}) \le Cn, \tag{1}$$

where $\mathcal{P}$ is an actual cycle packing and all uncovered original edges are counted as singletons. A component-rank normalization and the identity

$$C + 1 - \sigma = 12$$

then give Theorem 1.1. Keeping the singleton weight throughout the proof is essential: replacing weighted completion costs by unweighted ones at intermediate stages loses the final coefficient.

The quantitative proof has three parts. First, a mixed cycle-packing estimate with constants $43/10$ and 3 is combined with a short-circumference base and finite depth-first-search chains. Second, an indexed scale decomposition uses three final degree-reduction rates and preserves the exact assignment of every original edge. Third, a positive renewal potential pays the accumulated losses. Forty stopping-phase intervals determine different prefix thresholds; a positive second-order recurrence controls every possible number of layers.

The proof is computer-assisted. Two local interfaces require substantial finite certificates: the short-circumference base and the quantitative mixed-cover estimate. Their mathematical specifications and certification procedures are given in Sections 5, A, and B. The subsequent scale and cost calculations are displayed explicitly, with the rational data and the complete formal proofs supplied in the companion repository. Appendix D records the correspondence between the principal statements here and their Lean counterparts. The coefficient throughout this paper is exactly 12; no parameters from later numerical experiments are used.

2. Weighted packings and component-rank normalization

A *cycle packing* $\mathcal{P}$ is a family of pairwise edge-disjoint simple cycles in G. Let $p(\mathcal{P})$ be its number of cycles and let $q(\mathcal{P})$ be the number of edges not used by the cycles. Define

$$\mathrm{wt}_\sigma(\mathcal{P}) = p(\mathcal{P}) + \frac{23}{42}q(\mathcal{P}), \qquad z(\mathcal{P}) = 42p(\mathcal{P}) + 23q(\mathcal{P}). \tag{2}$$

The cycles together with the $q(\mathcal{P})$ remaining singleton edges form an exact decomposition of $E(G)$. We use the empty packing when G has no cycles. Since the possible integer costs form a nonempty subset of $\mathbb{N}$, a minimum-cost packing exists.

Lemma 2.1. *For a minimum-cost packing $\mathcal{P}$, the graph formed by its uncovered edges is a forest. Consequently $q(\mathcal{P}) \le r(G)$.*

Proof. If the uncovered graph contains a simple cycle of length $\ell \ge 3$, adding that cycle to the packing changes its integer cost by $42 - 23\ell < 0$, contrary to minimality. The uncovered graph F retains all n vertices, so $q(\mathcal{P}) = |E(F)| = n - c(F) \le n - c(G)$, since deletion of edges cannot merge components. □

The following finite amplification is the normalization used by the formal proof. In particular, no limiting argument is needed.

Lemma 2.2. *Suppose every finite simple graph H has a packing of integer cost at most $485\,|V(H)|$. Then every nonempty finite simple graph G has a packing of integer cost at most $485(|V(G)| - 1)$.*

Proof. Choose a vertex $v \in V(G)$ and form a graph W from 486 disjoint copies of G by identifying their copies of v. The resulting graph is simple and has $486(n-1)+1$ vertices. A simple cycle of W lies entirely in a single copy: to visit two copies it would have to pass through the common vertex twice before closing. Every singleton edge also belongs to one copy. Thus restriction of any packing of W gives a decomposition of each copy. If z is the minimum integer cost for G, every packing of W has cost at least $486z$.

The assumed order bound gives

$$486z \le 485\big(486(n-1)+1\big).$$

If $z > 485(n-1)$, integrality implies $z \ge 485(n-1)+1$; multiplying by 486 contradicts the preceding inequality. Therefore $z \le 485(n-1)$, as required. □

Proposition 2.3. *The weighted order bound (1) implies Theorem 1.1.*

Proof. Multiplying (1) by 42 gives the hypothesis of Lemma 2.2. Apply that lemma to each connected component of G and assemble the resulting packings. This gives a packing of integer cost at most $485r(G)$. Choose a minimum-cost packing $\mathcal{P}$ of G; then $z(\mathcal{P}) \le 485r(G)$ and Lemma 2.1 gives $q(\mathcal{P}) \le r(G)$. Hence

$$42\big(p(\mathcal{P}) + q(\mathcal{P})\big) = z(\mathcal{P}) + 19q(\mathcal{P}) \le 504r(G).$$

Dividing by 42 proves the assertion. Isolated components contribute zero to the rank and require no parts. If $n = 0$, the empty decomposition suffices. □

We shall prove a stronger form of (1), with an order-dependent factor $\phi(n) \le 1$. Every recursive call in that proof is on a graph with strictly fewer vertices; induced subgraphs, quotient graphs, and padded graphs are transported along explicit vertex bijections when necessary.

3. Classical inputs and their formal proofs

We record two inputs whose formal implementations are supplied in the artifact. The first uses the rotation-and-deletion argument attributed to Woodall [16]; see also [10, Theorem 2.7.1]. The second is a lower-tail estimate proved from finite product weights, the FKG inequality [8], and a Laplace-transform argument. These are implementations of classical methods, not claims of new extremal or correlation inequalities.

Lemma 3.1 (Erdős–Gallai cycle bound)**.** *Let G be a finite simple graph on n vertices and let $k \geq 2$ be an integer. If every simple cycle has length at most k, then*

$$2\,|E(G)| \leq k(n-1)$$

when $n \geq 1$. For $n = 0$ both sides of the corresponding natural-number statement are zero, with the predecessor interpreted as $n \dot{-} 1$.

Proof. We use strong induction on n. The edgeless case is immediate. Otherwise choose a longest path $P = v_0 \cdots v_\ell$, maximizing $d = \deg(v_0)$ among the starting degrees of all longest paths. Then $\ell \geq 1$. All neighbors of the first vertex of any longest path lie on that path, since an outside neighbor would extend it.

Define

$$W = \{v_i : 0 \leq i < \ell,\ v_0 v_{i+1} \in E(G)\}, \qquad X = \{v_j : 0 \leq j < \min\{k, \ell+1\}\}.$$

The predecessor operation is a bijection from $N(v_0)$ to W, so $|W| = d \geq 1$, and $v_\ell \notin W$. For each $v_i \in W$, the rotation

$$v_i v_{i-1} \cdots v_0 v_{i+1} \cdots v_\ell$$

is another longest path with the same vertex set. Consequently $\deg(v_i) \leq d$ and $N(v_i) \subseteq V(P)$.

A chord $v_0 v_j$ with $j \geq 2$ closes a prefix of P to a simple cycle of length $j+1$, whence $j < k$. The case $j = 1$ also has $j < k$. In particular, $v_i \in W$ implies $i + 1 < k$, and $W \subseteq X$. If $v_i \in W$ has a neighbor v_j with $j \geq k$, then $i + 1 < j$ and

$$v_0 \cdots v_i v_j v_{j-1} \cdots v_{i+1} v_0$$

is a simple cycle of length $j + 1 > k$. Thus every neighbor of W lies in X.

Let a count the edges inside W and let b count the edges between W and $X \setminus W$. The degree bounds and simplicity give

$$2a + b \leq d^2, \qquad b \leq d(|X| - d).$$

The $a + b$ edges removed with W therefore satisfy

$$2(a+b) \leq d\,|X| \leq dk.$$

The remaining graph has $n - d$ vertices, where $1 \leq n - d < n$, and retains the same bound on cycle lengths. Applying induction and adding the deletion cost yields

$$2\,|E(G)| \leq dk + k(n-d-1) = k(n-1).$$

No connectivity hypothesis is used. □

In Lean, the rotations and the degree-maximizing choice are implemented in

`Cycle12/Independent/PathRotation.lean`.

The set W and its neighborhood bounds are implemented in

`Cycle12/Independent/WoodallSet.lean`.

The counting step is implemented by splitting degree sums across W and its complement, so internal edges are counted exactly twice in the degree sum and exactly once in the deletion. The final strong induction is `Cycle12.Independent.erdosGallai_cycle`. Passing to a longest cycle gives the precise long-cycle interface used in the finite completion argument; its denominator is $n-1$.

Lemma 3.2 (Lower tail for a Bernoulli vertex sample)**.** *Let G have m edges, let $D \geq 0$, and suppose that $\deg(u)+\deg(v) \leq D+2$ for every edge uv. Independently select each vertex with probability $p \in [0,1]$, and let X count the edges with both endpoints selected. Put*

$$\mu = mp^2, \qquad v = mp^2 + mDp^3.$$

If $\mu > 0$ and $0 \leq t \leq \mu$, then

$$\mathbb{P}(X \leq \mu - t) \leq \exp\left(-\frac{t^2}{2v}\right).$$

If $\mu = 0$, the only required value is $t = 0$, for which the bound is the elementary probability bound by 1.

Proof. For a subset S of the vertex set, its probability mass is $w(S) = p^{|S|}(1-p)^{n-|S|}$. The identity

$$w(A)w(B) = w(A \cap B)w(A \cup B)$$

follows by adding the corresponding cardinalities and complement cardinalities. FKG gives positive association of nonnegative increasing functions. Shifting an increasing function by its value at the empty set extends this to real-valued increasing functions. Applying positive association to f and $M - g$, where g is decreasing and $M \geq \max g$, gives

$$\mathbb{E}(fg) \leq (\mathbb{E}f)(\mathbb{E}g) \tag{3}$$

for nonnegative increasing f and nonnegative decreasing g. Independence of observables using disjoint coordinates follows by induction on the finite ground set, conditioning on its last coordinate.

Write I_e for the indicator that edge e survives. For a fixed edge e, split

$$X = I_e + Y_e + Z_e,$$

where Y_e sums the indicators of the other edges sharing an endpoint with e, and Z_e sums all remaining indicators. There are at most D terms in Y_e. Coordinate independence gives, for $s \geq 0$,

$$\mathbb{E}(I_e e^{-sZ_e}) = p^2 \mathbb{E}(e^{-sZ_e}).$$

For each edge f contributing to Y_e, the increasing function $I_e I_f$ has expectation p^3; the formal estimate only needs the upper bound p^3. Applying (3) and summing gives

$$\mathbb{E}(I_e Y_e e^{-sZ_e}) \leq Dp^3 \mathbb{E}(e^{-sZ_e}).$$

The inequality $e^{-sY_e} \geq 1 - sY_e$, applied on the event $I_e = 1$, now gives

$$\mathbb{E}(I_e e^{-sX}) \geq e^{-s}(p^2 - sDp^3)\mathbb{E}(e^{-sZ_e}).$$

If $0 \leq s \leq (1+Dp)^{-1}$, the coefficient on the right is nonnegative. Since $Z_e \leq X$, putting $L(s) = \mathbb{E}(e^{-sX})$ yields

$$\mathbb{E}(I_e e^{-sX}) \geq e^{-s}(p^2 - sDp^3)L(s) \geq p^2\big(1 - s(1+Dp)\big)L(s).$$

For the second inequality, use $e^{-s} \geq 1-s$ and then discard the nonnegative term $s^2 Dp^3$. Summation over e gives

$$\mathbb{E}(Xe^{-sX}) \geq (\mu - vs)L(s).$$

Finite summation permits differentiation, with $L'(s) = -\mathbb{E}(Xe^{-sX})$. Hence

$$s \longmapsto e^{\mu s - vs^2/2} L(s)$$

is nonincreasing on $[0, \mu/v]$. Its value at 0 is 1, so $L(s) \le e^{-\mu s + v s^2/2}$ there. Finally, the exponential Markov inequality applied with $s = t/v$ proves the claim. The case $t = 0$ is handled directly by the probability bound by 1. □

The finite-law calculation is formalized first for a family of distinct two-element sets, with explicit dependency neighborhoods. The graph interface instantiates these sets with the endpoints of the original edges and uses the degree bound in the line graph. The statements `Cycle13.vertex_sample_edge_lower_tail` and `Cycle13.finite_graph_janson_restricted` retain the exact constants and the finite-sample conventions required later. The proof invokes Mathlib's FKG theorem and does not assume a separate Janson theorem.

4. Parameters and the renewal potential

All constants in this section are fixed rational numbers. Their exact values are listed in Appendix C; the symbols agree with the formal proof, except that c_0 denotes its constant `c`, B_0 its constant `batch`, and χ its constant `overlap`. In particular

$$C = \frac{485}{42}, \quad \sigma = \frac{23}{42}, \quad \kappa = \frac{999999}{1000000}, \quad \phi_0 = \frac{46220729}{50000000}, \quad B_{\mathrm{main}} = \frac{158387}{1000}.$$

The parameters satisfy

$$0 < s < \theta_2 \le \theta_1 \le \theta_0 < 1, \quad 0 < b < 1, \quad \rho = 1 + \delta + \varepsilon, \quad q_0 = \theta_1\theta_2, \quad q_0 w_A = w_0. \tag{4}$$

Put

$$h = \frac{3s - \rho - 2\alpha}{2} > 0, \qquad c(w) = c_0\, 2^{h(w - w_A)}.$$

This growing team scale will be used in Section 7. All logarithms in exponents and scale coordinates are to base 2; ln denotes the natural logarithm.

For $z > 0$, define

$$E(z) = \sum_{j \ge 0} 2^{-z\theta_2^{-j}}. \tag{5}$$

The series converges. Indeed $\theta_2^{-j} \ge 1 + j(\theta_2^{-1} - 1)$, which bounds it above by a geometric series. The losses at a scale $w \ge w_A$ are

$$F(w) = \left(\frac{1001}{1000} + (\rho - s) q_0 w\right) 2^{-(\alpha - s q_0) w}, \tag{6}$$

$$O(w) = \chi M E(\delta\theta_2^2 w), \qquad N(w) = M^2 F(w), \tag{7}$$

$$J(w) = \frac{M}{2} E(\varepsilon\theta_2^2 w), \qquad U(w) = M 2^{-(s-\lambda)w} + 2M^2 2^{-(s+b-\lambda-\rho)w}, \tag{8}$$

$$V(w) = \frac{\gamma M}{2c(w)} F(w). \tag{9}$$

The total positive loss used for coefficient 12 is

$$H(w) = O(w) + \kappa\big(N(w) + U(w)\big) + \frac{B_0 O(w) + J(w) + V(w)}{C}. \tag{10}$$

No cancellation between separately bounded positive losses is used in this potential.

Lemma 4.1. *Let $r_* = \varepsilon\theta_2^2$. Each loss in (7)–(9) is nonnegative and nonincreasing on $[w_A, \infty)$. Moreover*

$$H(y) \le 2^{-r_*(y-x)} H(x) \qquad (w_A \le x \le y). \tag{11}$$

Proof. The rational parameter inequalities imply

$$0 < r_* \le \min\{\delta\theta_2^2, \alpha - \theta_0 q_0, s - \lambda, s + b - \lambda - \rho, \alpha - \theta_0 q_0 + h\}.$$

Every summand of $E(aw)$ decays at least as fast as 2^{-aw}. For F, put $A = 1001/1000$ and $B = (\rho - s)q_0$. The certified entropy slope inequality at w_A is

$$B \le (\ln 2)(\theta_0 - s)q_0(A + Bw_A).$$

Since $B \ge 0$, its right-hand side increases with w. Differentiating $\ln F(w)$ therefore gives

$$\frac{F'(w)}{F(w)} = \frac{B}{A + Bw} - (\ln 2)(\alpha - sq_0) \le -(\ln 2)(\alpha - \theta_0 q_0).$$

Integrating gives the asserted exponential comparison for F and N. The additional factor $2^{-h(w-w_A)}$ gives the displayed rate for V. The two summands of U and the series defining O, J have the other rates listed above. Positive linear combinations preserve the comparison with the common rate r_*, proving (11). □

Define, for $x \ge bw_A$,

$$L(x) = \sum_{j \ge 1} H(x/b^j). \tag{12}$$

Lemma 4.1 implies convergence, by the same geometric comparison as for (5). Shifting the summation index gives

$$L(bw) - L(w) = H(w) \qquad (w \ge w_A). \tag{13}$$

The anchor certificate proves

$$L(bw_A) \le 1 - \phi_0. \tag{14}$$

Set

$$\Phi(x) = \begin{cases} \phi_0, & x < bw_A, \\ 1 - L(x), & x \ge bw_A, \end{cases} \qquad \phi(n) = \begin{cases} \phi_0, & n = 0, \\ \Phi(\log_2 n), & n > 0. \end{cases} \tag{15}$$

The separate definition at zero matches the value used in Lean.

Lemma 4.2. *The functions Φ and ϕ are nondecreasing, and $0 < \phi_0 \le \Phi(x) \le 1$. If $w \ge w_A$ and $n \ge 2^w$, then*

$$H(w) \le \phi(n) - \Phi(bw). \tag{16}$$

Proof. The summands in (12) are nonnegative and nonincreasing, so L is nonnegative and nonincreasing. At bw_A, inequality (14) prevents a downward jump in (15). This proves the bounds and monotonicity. Equation (13) gives $\Phi(w) - \Phi(bw) = H(w)$, and $\log_2 n \ge w$ gives (16). □

For later use define the batch and group fees

$$B(w) = \frac{2^{(\rho - s - \alpha)w}}{4c(w)} + \frac{1}{2}\, 2^{(\rho - s - \lambda)w}, \qquad g(w) = \frac{1}{2c(w)}. \tag{17}$$

They are nonnegative and nonincreasing. The endpoint certificates establish, for every $w \ge w_A$,

$$B(w) \le B_0, \qquad g(w) \le C\kappa\phi_0 \le C\kappa\Phi(bw). \tag{18}$$

All numerical assertions in this section are rational inequalities or certified inequalities involving powers of 2. Section 11 and Appendix C specify how they are checked.

5. Finite-degree completion

For real a and positive t, write $\mathcal{A}(a, t)$ for the assertion that every finite simple graph with n vertices and m edges has a packing of weight at most $an + m/t$. This section supplies the finite-degree cases of the main induction and the completion of the stopped remainder.

5.1. **The short-circumference starting point.**

Lemma 5.1 (Short-circumference certificate)**.** *If G has no simple cycle of length at least 7, then it has a packing $\mathcal{P}$ satisfying*

$$z(\mathcal{P}) \le 19r(G) + 6m.$$

The proof is a structural reduction followed by a finite certificate check; it is given in Appendix A. Both the reduction to the finite catalogue and the interpretation of its witnesses as actual original-edge packings are part of the formal proof.

Lemma 5.2. *For every integer $u \ge 7$,*

$$\mathcal{A}\left(\frac{19}{42} + \frac{1}{2}\sum_{j=8}^{u}\frac{1}{j}, u\right) \quad \textit{holds.} \tag{19}$$

Proof. Greedily delete simple cycles of length at least 7. Their number is at most one seventh of the number of deleted edges. On the remaining graph, Lemma 5.1, divided by 42, gives weight at most $19n/42 + m'/7$. Adding the deleted cycles proves $\mathcal{A}(19/42, 7)$. For the successor step from t to $t+1$, first remove cycles of length at least $t+1$. On the remaining graph, the Erdős–Gallai long-cycle edge bound [6] gives $2m' \le tn$. Consequently

$$\frac{m'}{t} - \frac{m'}{t+1} = \frac{m'}{t(t+1)} \le \frac{n}{2(t+1)}.$$

Apply $\mathcal{A}(a,t)$ to this remainder and pay the deleted cycles by their lengths at denominator $t+1$. This proves $\mathcal{A}(a + 1/(2(t+1)), t+1)$. Iterating that step from 7 to u gives (19). □

5.2. **The mixed cover.** Put $B_* = 43/10$, and define

$$D(n,m) = B_* n m^{1/3} + 3n^{3/2}, \qquad \psi(v) = \begin{cases} (53/200)v, & 0 \le v \le 1, \\ 1/2 - (47/200)v^{-1/3}, & v \ge 1. \end{cases}$$

The two formulas for ψ agree at 1. The function is nonnegative, nondecreasing and concave, and is at most $1/2$.

Lemma 5.3 (Mixed-cover certificate)**.** *Every graph of positive order has each of the following packings:*

$$p \le n\psi\left(\frac{m}{2B_*^3 n}\right), \qquad q \le D(n,m) + 2n; \tag{20}$$

$$p \le n/2, \qquad q \le D(n,m) + n. \tag{21}$$

The packings in the two lines need not be the same.

Proof. Use strong induction on the order. If $m \le D(n,m)$, the empty packing works. The greedy range is handled by the long-cycle estimate. A sparse cut is paid from the difference between the parent budget $D(n,m)$ and the two child budgets; cycles in the two induced sides are assembled and the cut edges are left as singletons. The perspective of the concave function ψ pays the cycle counts in (20). A sufficiently low-degree vertex is handled by deletion and the corresponding one-vertex budget inequality. The remaining graphs meet the degree and cut hypotheses of the routed-sampling construction. Its certificate covers *all* the remaining continuous parameters, including the unbounded scale range. Appendix B gives the construction, the precise coverage interface, and the finite checks closing this induction. □

5.3. **A certified depth-first-search step.** Consider a depth-first-search block with s new vertices and at most T boundary vertices, and let m' denote its assigned edges. The boundary is padded to T vertices when needed. Choose the local packings first, and then apply $\mathcal{A}(a,t)$ once to the graph of all uncovered original edges, retaining the original vertex set. Its intercept is therefore an. For a regular block, the fine packing contributes the following additional fee, after subtracting the block's assigned-edge budget m'/T:

$$(s+T)\psi\left(\frac{m'}{2B_*^3(s+T)}\right)+\frac{D(s+T,m')+2(s+T)}{t}-\frac{m'}{T}. \tag{22}$$

The alternative is to leave the block's edges for that common suffix estimate; its additional fee is $m'/t-m'/T$. An entire component with no boundary instead uses (21), giving fee

$$s/2+\frac{D(s,m')+s}{t}-\frac{m'}{T}. \tag{23}$$

These are different branches of the DFS decomposition, not interchangeable versions of the same packing.

For $g>0$ let $\Pi_p(g)=(1+g)^p g^{1-p}$. Write

$$\begin{aligned}\mathcal{K}(T,t,d,g)=(1+g)\left[\psi\left(\frac{d}{4B_*^3}\right)+\frac{2}{t}-\frac{d}{2T}\right]\\ +\frac{B_*}{t}\left(\frac{dT}{2}\right)^{1/3}\Pi_{4/3}(g)+\frac{3\sqrt{T}}{t}\Pi_{3/2}(g).\end{aligned} \tag{24}$$

The substitution $d=2m'/(s+T)$ and $g=T/s$ makes (22) equal to $s\mathcal{K}(T,t,d,g)$. The functions Π_p are convex for $p\ge 1$: they are perspectives of $x\mapsto x^p$. Thus, for fixed d, the maximum of (24) on a closed g-interval occurs at an endpoint, even when its linear coefficient is negative.

Lemma 5.4 (Finite DFS step)**.** *Let T,t be positive integers with $T\ge 2^{20}$, and let $\beta_0,K>0$. Suppose*

$$\begin{aligned}&(1+1/\beta_0)/2\le K,\\ &\mathcal{K}(T,t,d,g)\le K\quad (d>t),\quad g\in\{(2\beta_0+2^{-20})^{-1},\beta_0^{-1}\},\\ &s/2+\big(D(s,m')+s\big)/t-m'/T\le Ks\quad (0<s\le\beta_0T+1,\ m'\ge 0).\end{aligned}$$

Then $\mathcal{A}(a,t)$ implies $\mathcal{A}(a+K,T)$.

Proof. First remove simple cycles of length at least T. If h edges are removed, these cycles cost at most h/T. The remaining graph, on the same vertex set, has $m-h$ edges and no cycle of length at least T; this is the circumference hypothesis required by the DFS block construction. The rounded DFS construction assigns each remaining edge to exactly one block. Its regular blocks have T/s between the two displayed endpoints; the rounding error is bounded by $1/T\le 2^{-20}$. For $d\le t$, the direct suffix branch is paid by $(1+1/\beta_0)/2\le K$. For $d>t$, convexity and the two endpoint hypotheses pay (22). Boundary-free components have at most β_0T+1 vertices and are paid by the last hypothesis. Sum over the disjoint sets of new vertices. The assigned edge counts sum to $m-h$. Apply the suffix packing once to the actual residual graph, so its intercept is charged once per original vertex. This gives weight at most $(a+K)n+(m-h)/T$ on the remainder. Restoring the initially removed cycles adds at most h/T, yielding $(a+K)n+m/T$. The original-edge assembly is proved by `Cycle13.weightedAffine_dfs_step`; the scalar hypotheses are supplied in `Cycle12.FiniteComponentPayment`. □

The endpoint hypotheses in Lemma 5.4 are verified on whole real density intervals. In the high-density branch, $t\ge 4B_*^3=79507/250$. After substitution, the nonlinear density expression has the form $Ax^{1/3}-Bx^{-1/3}-x/2$; a certified tangent and derivative sign pay its entire domain.

Rational lower and upper bounds for A, B and $\sqrt{\cdot}$ are established by cubing or squaring nonnegative rational endpoints. The component branch has $B = 0$ and also includes zero density. No finite sampling of possible graph densities is used.

There are 83 finite chains. A chain begins with (19), uses a certified upper bound on its harmonic intercept, and applies Lemma 5.4 a finite number of times. If its final affine bound is $\mathcal{A}(a, T)$ and its degree cap is d_0, then

$$2m \le d_0 n \quad \Longrightarrow \quad \mathrm{wt}_\sigma(\mathcal{P}) \le \left(a + \frac{d_0}{2T}\right) n. \tag{25}$$

All T and d_0 in the formal certificates are integers; bounds such as $2^x \le d_0$ are certified before (25) is applied. The chains and their exact fees are tabulated in the companion data.

Proposition 5.5 (Finite potential base)**.** *If $2m \le 2^{B_{\mathrm{main}}} n$, then G has a packing of weight at most $C\phi(n)n$.*

Proof. The 45 base intervals cover the required order range, with their shared endpoints included. On each interval choose its certified finite chain. Use the simple-graph estimate $2m \le n^2$ together with the assumed cap to establish the chain's integer degree cap. The resulting constant in (25) is at most the lower endpoint value of $C\phi(n)$. Monotonicity from Lemma 4.2 pays the entire interval. Beyond the last interval the main finite chain and the same monotonicity apply. The zero-order case has no edges. The endpoint payments and the interval-coverage proof are `Cycle12.FinitePotentialPayments` and `Cycle12.FinitePotentialBase12`. □

6. An exact decomposition into scales

The large-order proof uses the scale decomposition of [14], with explicit parameters, a parity correction, and three final reduction rates. We specify which edges and which vertex multiplicities are counted.

A supported piece consists of a graph on the original vertex set together with a vertex support containing both endpoints of every one of its edges. Its mass is the size of that support. Different pieces may share vertices; their edge sets must be disjoint. The mass of a family is the sum of its support sizes, including multiplicity.

Lemma 6.1 (One scale)**.** *At logarithmic scale w, set*

$$D = 2^w, \quad \ell = 2^{(1+\varepsilon)w}, \quad L_0 = 2^{\rho w}, \quad h_0 = 2^{sw}.$$

Under the parameter inequalities at the applicable scale, a piece with average degree at most D can be partitioned into long cycles, even expanding cores, and a new family of supported remainder pieces. The following bounds hold per unit parent mass:

- *(i) the number of long cycles is at most $D/(2\ell)$;*
- *(ii) the sum of box orders is at most $1 + 2\ell/(L_0 - \ell - 2)$, and each box has order at most L_0;*
- *(iii) cores in a box have disjoint vertex supports, orders greater than h_0, and cut expansion at least $(1023/1024)h_0$; every core degree is even;*
- *(iv) the next average-degree bound is at most $h_0(1001/1000 + \log_2(L_0/h_0))$.*

Proof. First remove actual edge-disjoint cycles longer than the prescribed threshold; their lengths pay (i). On the cycle-free remainder, a DFS partition with integer parameters $\lfloor \ell \rfloor$ and $\lfloor (L_0 - \lfloor \ell \rfloor)/2 \rfloor$ gives the boxes in (ii). In particular, the denominator includes both rounding units. Within each box, repeatedly cut a side of insufficient edge expansion. The entropy potential pays the cut edges and gives the logarithmic term in (iv). The surviving induced cores have disjoint supports. For each core, choose a forest whose odd-degree vertices are exactly the odd-degree vertices of the core, and move that forest into the remainder. Subtraction makes the core even. The cut bound for the forest and the certified scale inequalities leave the expansion in (iii); its at most one edge

per vertex is absorbed by the 1001/1000 term in (iv). The remainder children are defined *after* this move. Each operation is a partition of original edges, so composition preserves the partition. The conditions $\ell + 2 < L_0$, $h_0 > 4096$, $h_0 \le L_0$, and $2048L_0 < h_0^2$ are checked on the complete scale ranges, not assumed from asymptotic notation. □

Let $X = \log_2 n \ge w_A$. Choose $k \ge 0$ so that

$$w_A \le W := \theta_2^k X < w_A/\theta_2, \qquad m_0 = k + 3. \tag{26}$$

(The symbol m_0 counts stages, not edges.) Define

$$w_i = \begin{cases} \theta_2^i X, & 0 \le i \le k+1, \\ \theta_1\theta_2^{k+1} X, & i = k+2, \\ \theta_0\theta_1\theta_2^{k+1} X, & i = k+3. \end{cases} \tag{27}$$

Apply Lemma 6.1 at stages $0, \dots, k+2$. The entropy inequalities for the three transitions give average degree at most $2^{w_{i+1}}$ in every child piece. The last processed scale is at least $q_0 w_A = w_0$. Let S_i be the total core mass at stage i.

For each prefix, the long cycles removed so far, the cores extracted so far, and the *whole* current frontier form an exact partition of $E(G)$. This prefix identity permits both operations used later: replacing one selected core layer together with its descendant frontier, or stopping before the last two stages. It would not suffice to retain only the sparse edges at the end of all stages.

The overlap factor in (ii) is at most $1 + \chi 2^{-\delta w}$. Products of these factors are bounded by $\prod_j (1 + x_j) \le (1 - \sum_j x_j)^{-1}$ when $x_j \ge 0$ and $\sum_j x_j < 1$. The certified endpoint inequalities give a uniform mass factor M. For a phase lower endpoint $a \le W$, define

$$M_1(a) = \frac{1}{1 - \chi(2^{-\delta q_0 a} + E(\delta\theta_2 a))}, \qquad M_2(a) = \frac{1}{1 - \chi E(\delta\theta_2 a)}, \qquad M_3(a) = \frac{1}{1 - \chi E(\delta a)}. \tag{28}$$

All denominators are positive. In particular

$$S_{m_0-1} \le M_1(a) n, \qquad S_{m_0-2} \le M_2(a) n. \tag{29}$$

Lemma 6.2 (Stopped remainder)**.** *For $a \le W \le d$, stop the actual partition before stage $k + 1 = m_0 - 2$. Merge its frontier pieces on the original vertex set and call the resulting graph H. Then*

$$2\,|E(H)| \le M_3(a) 2^{\theta_2 d} n.$$

The long cycles retained before this stop number at most

$$J_{\mathrm{tr}}(a) n, \qquad J_{\mathrm{tr}}(a) = \tfrac{1}{2} M_3(a) E(\varepsilon a). \tag{30}$$

Proof. The frontier average-degree bound is $2^{w_{k+1}} = 2^{\theta_2 W}$ per unit support mass. Its support mass is at most $M_3(a)n$. The pieces have disjoint edges, so summing their degree bounds proves the first inequality after the merge. Sum the long-cycle estimates from Lemma 6.1(i) over stages $0, \dots, k$; their scales are bounded below in reverse order by $a\theta_2^{-j}$. The same mass bound gives (30). □

Every retained even core also has a completion packing of weight at most $(1 + C\eta)$ times its support size, provided the weighted order bound is available for strictly smaller graphs. The completion construction uses an edge bank, actual paths, and a smaller-order remainder. We state the more precise version, including the extra credit needed for folding, next.

7. Routing, completion, and original-edge ownership

This section records the graph-theoretic interfaces used by the physical fold. They strengthen the routing and completion framework of [14] quantitatively and keep its original-edge interpretation. For a graph H, cut expansion at least h means

$$e_H(Z, V(H) \setminus Z) \geq h \min\{|Z|, |V(H) \setminus Z|\} \quad (Z \subseteq V(H)).$$

For a supported core, cuts are taken inside its support.

7.1. Two edge banks and smaller-order completion.

An even core at scale $w \geq w_0$ has expansion at least $(1023/1024)2^{sw}$. A biased edge partition, with bias $511/512$ and deviation $1/1024$, splits it into disjoint routing and completion banks of expansions at least

$$\frac{1044483}{1048576}\, 2^{sw}, \qquad \frac{1023}{1048576}\, 2^{sw}, \tag{31}$$

respectively. The cut-probability bound and its logarithmic scale gate are certified uniformly for core orders at most $2^{\rho w}$. There is no second parity correction: the cores were already corrected in Lemma 6.1.

For completion, the sampling-and-closing construction produces at most one cycle per core vertex and leaves a graph supported on fewer than η times as many vertices. Apply the smaller-order induction hypothesis to this induced residual graph and extend the packing by the constructed cycles. Its weight is at most $(1 + C\eta)$ times the core order. The strict support inequality, rather than a comparison of average degrees, justifies this induction call.

A fold also leaves high-degree external centres attached to a core. The following is the precise credit needed in this case.

Lemma 7.1 (High-centre completion)**.** *Let a graph be supported on $W = U \cup Z$, with $U \cap Z = \varnothing$ and $2 \leq |W|$. Suppose $w \geq w_A$, $2^{sw} \leq |U| \leq 2^{\rho w}$, the graph has cut expansion on U at least the second quantity in (31), and every external vertex has even degree and at least d neighbours in U. Suppose also*

$$a_w = 2^{\alpha w}, \quad \deg(v, Z) \leq a_w \ (v \in U), \quad \sqrt{2a_w |U|} \leq d \leq \sqrt{2a_w |U|} + 1.$$

If the weighted order bound with constant $K \geq 0$ holds for all graphs of strictly smaller order than $|W|$, then the graph has a packing with

$$\mathrm{wt}_\sigma(\mathcal{P}) + d/2 + 2a_w \leq (1 + \sigma + K\eta + 1/2048)\, |U|\,. \tag{32}$$

Proof. Put $q = |Z|$ and $v = |U|$. Double-counting edges from Z to U gives $qd \leq a_w v$, and $d^2 \geq 2a_w v$ gives $q \leq d/2$. The scale inequalities give the stronger rounding budget

$$q + d/2 + 2a_w \leq v/2048.$$

They also allow the core expansion to extend across the high centres with the effective expansion required for completion. The completion construction then gives weight at most

$$(v + q) + (K\eta + \sigma)v.$$

Only vertices of U need a parity charge: vertices of Z already have even degree. Add $d/2 + 2a_w$ and use the rounding budget to obtain (32). The proof applies on the actual support W; isolated padding is removed by a cost-preserving embedding. All residual induction calls have fewer vertices than W. □

7.2. **A single routing system for all requests.** At an active scale put

$$Q(w) = \lceil c(w)2^{\alpha w} \rceil . \tag{33}$$

Requests consist of an ordered pair of endpoints, a team index, a set of forbidden interior vertices, and a nonnegative vertex weight. Endpoint load counts occurrences, so repeated endpoints are retained.

Lemma 7.2 (Grouped weighted routing)**.** *In an even core at scale $w \geq w_A$, suppose that each vertex has endpoint load at most $2^{\alpha w}$, each team contains at most $Q(w)$ requests, and each forbidden set has size at most $2Q(w)$. There are paths in its routing bank for all requests such that distinct paths are edge-disjoint; paths in one team have disjoint interiors; each interior avoids its forbidden set; and for every request with weight function ω,*

$$\sum_{v \text{ in its interior}} \omega(v) \leq \frac{\gamma}{Q(w)} \sum_{v \in U} \omega(v). \tag{34}$$

Their lengths satisfy the scale-dependent bound `globalRoutingLength` *used in the capacity certificates.*

Proof. For the large-ratio case, write v for core order, h for the routing expansion, $B = Q(w)$ for team capacity and k_0 for endpoint capacity. Put $q = 1/c_0$, $a = a_{\mathrm{prune}}$, $R = 7519120361/500000000$ and $L = 30589490361/5000000000$. The exact rational parameters satisfy

$$1 + q + aq < 2a^2(q+2), \qquad 4(1+q)/a < R,$$
$$r_0 := 1 - a - 1/128 - 1/(R\gamma) > 0, \qquad 8/r_0 < (L - 1/32)^2.$$

The scale inequalities give $k_0 \leq qB$ and $R^2L^2B^2v \leq h^3$. With $\ell = \left\lfloor L\sqrt{v/h} \right\rfloor$ this implies $RB\ell \leq h$.

Associate to a candidate path its edge set and the marked interior vertices (team, v). These are resources in $E(G) \sqcup (\mathcal{T} \times V(G))$. A path of length at most ℓ uses at most 2ℓ resources. To apply the Aharoni–Haxell selection criterion [1], it suffices to show that for every nonempty request subfamily I and every resource set Z of size less than $4\ell\,|I|$, some request in I has an admissible path avoiding Z.

Write F for the edge resources of Z and Y for its marked-vertex resources. Put $D = 4\ell\,|I|$ and $x = |F|\,/D$. Adaptive pruning chooses $z, \tau > 0$ with

$$z + \tau = a, \qquad \frac{qx}{z} + \frac{1-x}{\tau} = \frac{1+q}{a}, \qquad \frac{qax}{2(1+q)} < \frac{z^2}{1+z}.$$

After deletion of F, pruning discards at most $|F|\,/(zh)$ vertices and leaves expansion $(1-z)h$. Endpoint and team loads imply

$$\frac{k_0\,|F|}{zh} + \frac{B\,|Y|}{\tau h} < |I|\,.$$

Thus a request remains whose endpoints survive and whose team has at most τh marked interior obstructions. Delete those marks together with its hard ports, and construct a short path in the surviving expanding core. The short-path fan and averaging argument gives the additional weight bound $(R\gamma\ell/h)\sum\omega \leq (\gamma/B)\sum\omega$. The retained expansion r_0h gives the required path length ℓ. This proves the resource-avoidance condition. Aharoni–Haxell then selects all paths simultaneously with disjoint resources, giving both forms of disjointness in the statement. The formal selection lemma keeps repeated endpoint occurrences as separate requests.

The actual integer $Q(w)$ is used throughout. Its rounding inequalities prove the endpoint, team, hard-port and cubic-capacity gates at w_A and extend them to the full scale half-line.

There are two size regimes. If core order divided by routing-bank expansion is at least 4096, use the grouped weighted router directly. Otherwise use the comparison bank and its bounded-ratio

routing theorem. The comparison ratio remains at least 4 after the original parity correction. Both regimes have the same conclusion (34). Thus no intermediate range of core orders is omitted.

The completion machinery additionally uses an indexed reservoir router. Its explicit gates are $z \geq 8192$, $8192 \ln v \leq z$, $2112\,|I| \leq z^2$, minimum degree at least $15z/16$, expansion at least $z/30$, and endpoint load at most $z/64$. A balanced bank split leaves one expanding bank to connect endpoint occurrences to a weak clique immersion in the other. The clique-immersion theorem of [9] and the terminal-pairing construction of [11] then produce actual edge-disjoint paths. Only edge-disjointness is needed from this reservoir theorem; it does not assert disjoint interiors. All these graph constructions, including their integer capacities, are proved in the formal source, not supplied by numerical gate checks alone. □

The key bookkeeping object for applying Lemma 7.2 is an *original endpoint certificate*. It assigns to each occurrence of an endpoint of a request an incidence (e, v) of an original edge, where v is that endpoint. The assignment is injective. If the original edges used by two families are disjoint, their certificates combine. Reordering the incoming endpoints within the same owner core preserves the certificate.

Three families are combined before routing: low two-edge paths, requests from paired matching edges, and the switches of quotient cycles. Their original edge sets are disjoint. The combined injection implies that the *total* endpoint load at v is bounded by its degree in the assigned descendant graph. Active-block preparation bounds this degree by $2^{\alpha w}$. Bounding the three families separately and adding those bounds would not establish the required load.

The combined team construction also preserves the $Q(w)$ capacity and $2Q(w)$ hard-port limit. Apply Lemma 7.2 once to each selected core, and use those same paths in all subsequent closures. Low groups and paired matching groups close into actual simple cycles. Quotient cycles are lifted through their switches, with the soft charge (34). The untouched completion bank remains available; the core remainder deletes exactly the paths actually used. These ownership rules are the reason that the four families in the next section have both an exact edge cover and pairwise disjointness.

8. Folding a layer and paying its cost

We work under the strong-induction hypothesis that every graph on fewer than n vertices has a packing of weight at most $C\phi(v)v$. At an active stage $0 \leq i \leq k$, put $w = w_i \geq w_A$ and

$$S = S_i, \qquad T = \sum_{j=0}^{i+2} S_j, \qquad f = \Phi(bw).$$

Descendant pieces are grouped inside their actual DFS boxes into batches. Write D for total batch order and A for the total size of the active blocks that are contracted. The graph construction proves

$$S - (N(w) + U(w))n \leq A \leq S + O(w)n, \qquad D \leq (1 + O(w))n. \tag{35}$$

Here N pays the small or unusable descendants, U the discarded boundary and grouping terms, and O the overlapping support multiplicity. Each batch has order at most 2^{bw}. Its active blocks are pairwise vertex-disjoint and belong to uniquely determined parent cores.

Lemma 8.1 (Physical fold)**.** *There is an actual packing of the whole graph G of weight at most*

$$\begin{aligned}(1 + C\eta)T + \sigma S + S/2048 + (B(w) + g(w))A \\ + Cf(D - \kappa A) + (J(w) + V(w))n.\end{aligned} \tag{36}$$

Proof. For each batch, retain one original representative of every quotient edge; classify the original edges lost by contraction as internal pairs or matching requests. This produces a simple quotient. An active block saves at least κ times its order in the quotient order, so the summed quotient

orders are at most $D - \kappa A$. Each quotient has order at most $2^{bw} \le n^b < n$, since $n > 1$ and $b < 1$. Apply the induction hypothesis and monotonicity of ϕ to obtain quotient packing weight at most $Cf(D - \kappa A)$.

Use the joint routing construction of Section 7 on each selected core. The single original-edge-endpoint injection covers low pairs, paired matching edges, and switches in quotient cycles simultaneously. The resulting four families are: completed core graphs, closed low groups, closed matching groups, and lifted quotient packings. Uncovered quotient edges keep their chosen original singleton representatives. Unused routing paths are not deleted from the core.

The completed core and low-group families have combined weight at most

$$(1 + \sigma + C\eta + 1/2048)S + (\text{prepared pair count})/Q(w).$$

This bound uses the high-centre credit of Lemma 7.1 before any mass upper bound is substituted. The matching groups and the prepared-pair fee together are at most $(B(w) + g(w))A$, and the soft routing charge for lifted quotient cycles is at most $V(w)n$. The quotient packing contributes the term already obtained above.

The four families are pairwise edge-disjoint and their union is exactly the selected cores and the full descendant frontier. To see the importance of both assertions, separate original non-routing edges from the paths deleted during routing. Every original edge has exactly one of the four owners by the quotient classification. Every deleted path has exactly its request owner; joint routing makes different owners edge-disjoint. An edge cannot belong to both classes because the core remainders were defined by deleting precisely the selected paths. This is the partition proved in `ActualFoldDisjoint12` and `ActualFoldCover12`.

Finally insert the other prefix cores, of total mass $T - S$, at cost $(1 + C\eta)(T - S)$, and the prefix long cycles at cost at most $J(w)n$. The prefix partition from Section 6 now covers the whole original graph. Adding the displayed costs proves (36). □

Define the coefficient of the selected core mass by

$$\Gamma(P, w) = (1 + C\eta)P + \sigma + B(w) + g(w) + \frac{1}{2048} - C\kappa\Phi(bw). \tag{37}$$

Lemma 8.2 (Fold ledger)**.** *If $T \le PS$, then the right-hand side of (36) is at most*

$$Cfn + \Gamma(P, w)S + CH(w)n. \tag{38}$$

Proof. Write the terms involving A as $BA - (C\kappa f - g)A$. The second coefficient is nonnegative by (18). Consequently the *lower* bound on A in (35) is used in its negative term, and the upper bound is used only in BA. The resulting error coefficient of n is

$$CfO + (C\kappa f - g)(N + U) + BO + J + V.$$

All losses are nonnegative, $0 < f \le 1$, $g \ge 0$, and $B \le B_0$, so this is at most $CO + C\kappa(N + U) + B_0O + J + V = CH$. The coefficient of S is exactly (37). This proves (38). In particular, one does not introduce an extra gO term by applying an upper bound to the negative group term. □

For each of the 40 phase intervals $[a, d]$, the certificates supply positive thresholds P_t for all $t \ge 3$. They prove

$$\Gamma(P_t, w) \le 0 \quad \text{whenever} \quad w \ge a\theta_2^{-(t-3)}. \tag{39}$$

There are explicit thresholds for $3 \le t \le 16$ and a constant threshold for $t \ge 16$. The scalar proof uses the monotonicity of B, g, Φ; the constant tail therefore requires only its first scale endpoint.

If i is a stage with $S_i > 0$ and

$$\sum_{j=0}^{i+2} S_j \le P_{m_0-i}S_i, \tag{40}$$

then $t = m_0 - i$ and (27) give $w_i = W\theta_2^{-(t-3)} \geq a\theta_2^{-(t-3)}$. Lemmas 8.1 and 8.2, followed by (39) and (16), give

$$\mathrm{wt}_\sigma(\mathcal{P}) \leq Cfn + CH(w_i)n \leq C\phi(n)n.$$

This completes every branch in which an active fold exists.

9. The no-fold branch and the infinite tail

Fix the phase interval $[a, d]$ containing W. Alongside P_t, its certificate gives a positive summable sequence $(u_t)_{t\geq 1}$ such that

$$u_1 \geq M_1(a), \qquad u_2 \geq M_2(a), \qquad \sum_{s\geq t-2} u_s \leq P_t u_t \quad (t \geq 3). \tag{41}$$

The initial terms are rational. The continuation is the positive recurrence

$$u_{t+2} = (1-r)u_{t+1} + r(1-r)u_t \quad (t \geq 15), \qquad r = \frac{88331750161}{10^{11}}. \tag{42}$$

Since $0 < r < 1$, the recurrence is summable. If $\beta = r(1-r)$ and $Q = 1/[r^2(1-r)]$, then

$$\sum_{s\geq 15} u_s = Q(\beta u_{15} + (1-r)u_{16}). \tag{43}$$

This follows by summing the recurrence. Its positive characteristic root is less than 1, or equivalently the same conclusion follows by the geometric majorant used in the formal tail proof. The certificates check the finite prefix of (41) and the recurrence conditions that extend it to every t; in particular $Q < P_{16}$.

Lemma 9.1 (No-fold mass bound)**.** *If no active stage with positive core mass satisfies* (40)*, then*

$$S_i \leq nu_{m_0-i} \quad (0 \leq i < m_0).$$

In particular the cores retained before the terminal stop have total mass at most $n\sum_{s=3}^{m_0} u_s$.

Proof. If the conclusion fails, take the maximum $R = \max_i S_i/(nu_{m_0-i}) > 1$. The last two stages cannot attain this maximum, by (29) and (41). At a maximizing active stage i, set $t = m_0 - i \geq 3$. Then

$$\sum_{j=0}^{i+2} S_j \leq Rn \sum_{s=t-2}^{m_0} u_s \leq RnP_t u_t = P_t S_i.$$

Here $S_i > 0$, so this is an active fold, a contradiction. The retained stages are $0, \ldots, m_0 - 3$, giving the last assertion. □

Write

$$T_u = \sum_{s\geq 3} u_s, \qquad R_m = \sum_{s\geq m+1} u_s.$$

For the actual stopped graph H in Lemma 6.2, the phase certificate chooses one finite chain with fee $f_{a,d}$ and integer cap at least $M_3(a)2^{\theta_2 d}$. Thus H has a packing of weight at most $f_{a,d}n$. Packing the retained cores at cost $1 + C\eta$ per unit mass and adding the retained long cycles gives

$$\frac{\mathrm{wt}_\sigma(\mathcal{P})}{n} \leq f_{a,d} + (1 + C\eta)(T_u - R_{m_0}) + J_{\mathrm{tr}}(a). \tag{44}$$

The negative tail $-R_{m_0}$ is kept throughout; it is an unused portion of a summable majorant, not a negative physical graph cost.

Lemma 9.2 (Terminal payment)**.** *For every phase and every integer* $m \geq 3$*,*

$$f_{a,d} + (1 + C\eta)(T_u - R_m) + J_{\mathrm{tr}}(a) \leq C\Phi(a\theta_2^{-(m-3)}). \tag{45}$$

Proof. The phase data first certify

$$f_{a,d} + (1 + C\eta)T_u + J_{\mathrm{tr}}(a) \le C.$$

For $3 \le m < 15$, they directly certify

$$CL(a\theta_2^{-(m-3)}) + f_{a,d} + (1 + C\eta)T_u + J_{\mathrm{tr}}(a) \le C + (1 + C\eta)R_m.$$

These are finite rational interval certificates for the positive series in Section 4.

For all later m, a stronger inequality suffices:

$$CL(a\theta_2^{-(m-3)}) \le (1 + C\eta)R_m. \tag{46}$$

It is checked separately at $m = 15$ and $m = 16$. Positivity in (42) gives $R_{m+2} \ge \beta R_m$. On the other hand Lemma 4.1, applied to every summand of L, gives

$$L(x/\theta_2^2) \le 2^{-(r_*/b)x(\theta_2^{-2}-1)}L(x).$$

The certificate at $x = a\theta_2^{-12}$ bounds the exponential factor by β. The factor only decreases as x increases. Induction separately on the two parities therefore proves (46) for every $m \ge 15$. Combine it with the total-budget inequality and use $\Phi(x) = 1 - L(x)$ on this range to get (45). □

The 40 closed phase intervals cover $[w_A, w_A/\theta_2]$. At a shared endpoint either adjacent certificate can be used. For the actual phase, $X = W\theta_2^{-(m_0-3)} \ge a\theta_2^{-(m_0-3)}$. Equations (44) and (45), and monotonicity of Φ, show that the no-fold branch also has weight at most $C\phi(n)n$.

10. Completion of the proof

Theorem 10.1 (Weighted potential bound)**.** *Every finite simple graph on n vertices has a packing of weight at most $C\phi(n)n$, where $C = 485/42$ and ϕ is defined in (15).*

Proof. Use strong induction on n. If $2m \le 2^{B_{\mathrm{main}}}n$, apply Proposition 5.5; this includes $n = 0$. Otherwise the simple-graph inequality $2m \le n^2$ implies $n > 2^{B_{\mathrm{main}}}$, and hence $X = \log_2 n \ge w_A$. Construct the actual scale decomposition of Section 6. All completion and quotient calls use graphs of strictly smaller order, so the induction hypothesis supplies their packings.

If an active stage satisfies (40), apply the physical fold and its scalar payment in Section 8. Otherwise apply the no-fold construction and terminal payment in Section 9. In both cases the packing is on the original graph and has the claimed weight. This completes the induction. □

Proof of Theorem 1.1. Since $\phi(n) \le 1$, Theorem 10.1 implies (1). Proposition 2.3 then gives $\mathrm{ce}(G) \le 12r(G)$, including the empty graph and isolated components. □

11. Formal statement and verification artifact

The formalization uses Lean 4 [5] and Mathlib [15]. The declaration

`Cycle12.coefficient12`

proves:

> For every natural number n and simple graph G on $\mathrm{Fin}\, n$, there exists $k \in \mathbb{N}$ such that
> `Cycle13.EdgeDecomposition G k`
> and `k ≤ 12 * (n - Cycle13.componentCount G)`.

The edge-decomposition predicate consists of a family of k edge sets, each a simple cycle edge set or a singleton original edge, with pairwise disjointness and union equal to G's edge set. The component count is the cardinality of the type of all connected components. An additional inspection example unfolds these predicates and writes the arithmetic with `Nat.le`, `Nat.mul`, and `Nat.sub`. This checks the intended statement, including isolated components, without any additional graph hypothesis.

The supplied closure contains 10 554 project modules. The environment is Lean 4.34.1 and Mathlib revision

`d13f23b723b8a846827a245b89c10fc7d3f11612`.

All project proof bodies in this release were recompiled from source in this layout. After a temporary-storage reset, some lost object files were recovered from historical caches only when their bytes matched the hashes already recorded by this replacement build. Their prerequisite objects were also required to match that recorded build; other objects were regenerated. This recovery does not substitute a historical compilation for a current-source compilation. The journal and recovery evidence distinguish these operations. Preliminary replacement-interface builds belong to the same run and use matching source, dependency and object hashes. Fixed external Lean/Mathlib objects were reused; their sources were not recompiled. Where import-header snapshots were used, an ordinary and a snapshot-assisted compilation first produced byte-identical objects for a current source. Every subsequent source body was still recompiled. The final journal has a successful record for every retained module.

A traversal of the actual declaration types and proof bodies from the final theorem found 1 068 913 dependencies, of which 999 202 belong to the retained project modules. These include generated certificate lemmas, definitions, constructors and recursors; they are not a count of separately authored mathematical ideas. The retained modules contain no new axioms. The only axioms in the final closure are

`propext`, `Classical.choice`, `Quot.sound`.

No dependency is a `sorryAx`, unsafe declaration or partial declaration.

The historical declaration namespace `Cycle13` is retained for compatibility. Current source paths are under `Cycle12/Foundations` and `Cycle12/Independent`; this name does not change the constant in `Cycle12.coefficient12`. Earlier compilation reports are historical evidence, not substitutes for checking the replacement proofs. The replacement graph bound follows Lemma 3.1; the sampling bound follows Lemma 3.2. Neither implementation imports `ErdosProblems` or `ProbMethodCombinatorics`. No empty-environment replay of the full declaration closure is claimed.

The full large dependency report is included as a gzip file, together with its uncompressed SHA256 digest. The reproducibility scripts check every source against the accepted manifest, rebuild the local closure in dependency order, and check the exact theorem. They do not accept a file's existence as evidence that it was compiled successfully. The repository documentation separates reproducing the full proof from checking the packaged integrity and inspecting an already built target.

The manuscript is a mathematical account of this formal proof, with its principal correspondence listed in Appendix D. Lean verifies the formal declarations, not the correctness of this prose or its bibliographic claims. The computational certificates in the source are part of the proof artifact. Their generators and exploratory numerical searches are not trusted proof oracles. This preprint does not claim independent peer review or an optimal coefficient.

Appendix A. The short-circumference reduction and catalogue

We describe the computer-assisted proof of Lemma 5.1 in enough detail to specify both its infinite reduction and its finite verification. For a nonempty connected graph the target is

$$z(\mathcal{P}) \le 19(n-1) + 6m. \tag{47}$$

A cut-vertex split is additive for this target: if its two sides meet only at the cut vertex, their $(n_i - 1)$ terms sum to $n - 1$. Thus strong induction reduces to a block that remains connected

after deletion of any one vertex. The small-order cases are checked directly. The corresponding result for circumference at most 5 is proved first by the same block induction.

If a minimal counterexample to (47) has circumference 6, choose a six-cycle and label its vertices cyclically by $0, \dots, 5$. The structural reduction uses ears between cycle vertices. An ear and the longer of the two cycle arcs cannot form a cycle of length at least 7. The allowed ear endpoints and the compatibility of two ears are checked with explicit path-splicing witnesses. Together with deletion of repeated attachments, this shows that the graph outside the chosen cycle is a set of isolated vertices together with at most one edge.

Here the induction reductions are essential. For example, two degree-two vertices with the same pair of neighbours can be deleted and their four edges restored as a four-cycle. Restoring costs 42, whereas the target allowance increases by $2 \cdot 19 + 4 \cdot 6 = 62$. The source also constructs the analogous reductions for two exterior edge components with common hubs and for three vertices with an alternating three-neighbour attachment. They leave at most one vertex of each two-neighbour type and at most two of each alternating three-neighbour type. These are reductions of actual proper induced graphs, not assumptions that the whole graph has small order.

The 11 possible attachment types, in their index order $0, \dots, 10$, are

$$\{0,2\}, \{1,3\}, \{2,4\}, \{3,5\}, \{0,4\}, \{1,5\}, \{0,3\}, \{1,4\}, \{2,5\}, \{0,2,4\}, \{1,3,5\}.$$

These are the neighbourhoods in `SixAttachmentModels`. Pairwise incompatible types have explicit simple long-cycle witnesses. All pairwise compatible type sets are contained in one of the following 14 families, using the type indices of that file:

$$\{0,3,6\}, \{1,3,6\}, \{0,4,6\}, \{1,4,6\}, \quad \{1,4,7\}, \{2,4,7\}, \{1,5,7\}, \{2,5,7\},$$
$$\{0,2,8\}, \{0,3,8\}, \{2,5,8\}, \{3,5,8\}, \quad \{0,2,4,9\}, \{1,3,5,10\}.$$

The first nine types have multiplicity at most 1, and the last two at most 2. If an exterior edge is present, only the first four families are possible after relabelling; its endpoints attach at opposite cycle vertices $0, 3$, with either or both of the two optional cross attachments.

The common model uses 16 slots: six cycle vertices, four pairs of attachment slots, and two slots for the optional exterior edge. At most 11 slots are active. The 12 three-type families have 8 multiplicity choices each, and the last two families have 24 choices each. There are also $4 \cdot 8 \cdot 4$ nonempty exterior-edge choices. Hence the catalogue has

$$12 \cdot 8 + 2 \cdot 24 + 4 \cdot 8 \cdot 4 = 272$$

parameter rows. The inverse index in `SixResidueParameterCoverage` proves that every valid tuple is in this list. All nine non-cycle chords of the six-cycle are optional, giving $272 \cdot 2^9 = 139\,264$ cases.

A certificate row is either a long-cycle word, or a list of pairwise edge-disjoint cycle words. It records a set of required chords, and in the packing case an upper bound h on the total number of chords. If b_0 is the number of base edges, v_0 the number of active vertices, c_0 the number of certified cycles and e_0 their total number of covered edges, the packing check is

$$42c_0 + 17(b_0 + h) \le 19(v_0 - 1) + 23e_0. \tag{48}$$

For any accepted chord set, $m' \le b_0 + h$, so (48) implies

$$42c_0 + 23(m' - e_0) \le 19(v_0 - 1) + 6m'.$$

Every cycle word is checked for length, distinct vertices, consecutive adjacencies and edge-disjointness. A long-cycle word excludes every chord superset containing its required chords. The finite check proves that each of the 512 chord sets in each parameter row is accepted by one of its certificates. All these checks are evaluated with `decide +kernel`.

Finally, actual outside vertices are enumerated in their attachment fibres. The multiplicity bounds give an injection into the designated slots; coverage gives a bijection to the active slots, and

the classification gives the complete graph equality, including the exterior edge in both orientations. The finite witness transfers through this bijection. The unused slots are isolated; removing them preserves the number of cycles, the number of uncovered edges and the integer cost. This proves (47). Summing over connected components proves Lemma 5.1, including isolated vertices and the graph of order zero.

Appendix B. Certification of the mixed-cover construction

This appendix specifies the hard case in Lemma 5.3. The certificate concludes the existence of actual packings; it is not merely a numerical upper bound for a hypothetical sample.

B.1. **Coordinates and fixed-size samples.** For $m > 0$ put

$$x = \frac{\sqrt{n}}{B_* m^{1/3}}, \quad s' = \frac{n^2}{2m} \geq 1, \quad W(x) = B_*^3 x^2, \quad v(x) = W(x)(1+3x).$$

Then

$$n = \frac{4B_*^3 (s')^2}{W(x)^3}, \qquad m = \frac{n^2}{2s'}, \qquad D(n,m)/m = v(x). \tag{49}$$

The irreducible range is $0 < x \leq 17/250$, $s' \geq 1$. Set

$$p_0(x) = v(x)\left(\frac{99}{200} + \frac{11}{100}v(x) + \frac{7}{100}v(x)^2\right), \qquad N = \lfloor p_0(x) n \rfloor, \quad p = N/n.$$

The sample is uniform among the N-element subsets of $V(G)$. In particular, the proof does not substitute independent vertex sampling for this fixed-cardinality law. For $P = p/W(x)$ and $P_0(x) = p_0(x)/W(x)$ the exact floor bounds are

$$P_0(x) - \frac{W(x)^2}{4B_*^3 (s')^2} \leq P \leq P_0(x).$$

Monotone endpoint bounds turn these into rational bounds on every parameter rectangle.

Let

$$H(x) = 103/250 + (351/200)x, \qquad D_0(x) = 1 - (33/125)W(x),$$

$$h_G = B_* m^{1/3} H(x)/D_0(x), \qquad z = p h_G.$$

Here $H(x)$ is a local coordinate function, unrelated to the loss $H(w)$ in Section 4. The graph is assumed to meet the sparse-cut irreducibility and minimum-degree inequalities left by the induction in Lemma 5.3.

A good sample U has size exactly N, certified lower neighbour degrees into U, cut expansion inside U, an anchoring inequality, at least $(97/100)p^2 m$ induced edges, and at most $D(n,m)$ edges incident with U. The anchoring inequality, for $X \subseteq V \setminus U$ and $Y \subseteq U$ with $2|Y| \leq |U|$, is

$$L|X| \leq e_G(X, U \setminus Y) + e_G(Y, U \setminus Y).$$

Some rectangles also require a local cut bound up to sample proportion $1/125$. The certified selection theorem bounds the union of six bad events: low degree, a failed global cut, failed anchoring, too few sample edges, too many incident edges, and the optional local-cut failure. Their rational upper bounds have sum strictly less than 1. Thus at least one actual N-element sample is good. The fixed-size moment, degree, marked-cut and anchor bounds and their parameter matches are proved before this union bound is used.

B.2. **From the sample to cycles.** A capacity certificate splits the induced sample edges into banks, constructs a weak clique immersion in the appropriate bank, and uses the remaining expansion to route the requests from the path-and-cycle decomposition of Lovász [13] applied to the complement. Weak immersion is sufficient: routed paths are edge-disjoint; branch vertices are not silently assumed to be avoided by all interiors. The indexed routing and cycle-closure lemmas account for endpoint occurrences and integer terminal capacities. The global-cut and additional-local-cut branches have separate capacity inequalities.

The output is an actual packing $\mathcal{C}$ in G with

$$p(\mathcal{C}) \le \frac{n-|U|}{2}, \qquad q(\mathcal{C}) \le D(n,m)+n-|U|\,. \tag{50}$$

The sample-size lower bound, including its floor, pays the fine cycle budget $n\psi(m/(2B_*^3 n))$. The second inequality in (50) pays both allowed residual-edge budgets, and the first is also at most $n/2$. Thus this hard case proves both conclusions of Lemma 5.3. The full induction may still choose different packings for its two conclusions.

B.3. **Continuous coverage and its soundness interface.** A rational rectangle is a tuple (x_-, x_+, s_-, s_+), where s_+ may be absent, denoting an unbounded ray. Its *soundness* statement quantifies over every actual graph whose coordinates lie in it and which satisfies the irreducibility and degree conditions, and concludes (20) and (21). A routed case consists of a sample certificate and a capacity certificate. Their common parameters—degree, expansion, anchoring and any optional local cut constant—must agree exactly. The theorem `rat_routed_case_sound` combines their graph-theoretic conclusions only after these equalities and domain inclusions are checked.

Rectangles are joined by exact rational interval chains. Horizontal joining checks that the x intervals cover and that each piece extends over the required s' interval. Vertical joining checks the converse. Both operations include closed endpoints, and an unbounded target ray must be covered by a chain that actually reaches an unbounded piece. These are proved soundness-preserving operations, rather than a sampled-grid argument. The final assembly is

$$[0,3/1000]\times[1,\infty) \quad\cup\quad [3/1000,17/250]\times[1,\infty),$$

with each rectangle further split into its certified scale pieces. Its final statement is

`Cycle12.CoverAssembly.Full.sound`.

Actual graphs have $x>0$; including $x=0$ in the interval certificate causes no division by zero in the graph application.

This closes the irreducible case and hence the strong induction. The certificate hierarchy contains the actual probability bounds and routing capacities, not just the final interval endpoints. Its rational decisions are checked in Lean, and its probabilistic and graph-theoretic soundness lemmas lie in the same axiom-audited dependency closure as the main theorem.

Appendix C. Exact parameters and certificate indices

The following values are read from `GlobalParameters12` and checked for exact equality with the documentary input `paper/data/global12.json`. Fractions in this table are exact. Large finite certificate entries remain in the Lean source; the supplementary JSON files provide their indices.

Parameter	Exact value
w_0	$\frac{10170985991133}{100000000000}$
s	$\frac{38082365177}{50000000000}$
θ_0	$\frac{2048519349}{2500000000}$

Parameter	Exact value
θ_2	$\frac{15961744051}{20000000000}$
α	$\frac{13908880001}{25000000000}$
λ	$\frac{3511913879}{10000000000}$
b	$\frac{3004140221}{4000000000}$
γ	$\frac{2460109537}{1562500000}$
δ	$\frac{3454748071}{50000000000}$
ε	$\frac{3093377987}{100000000000}$
c_0	$\frac{284161497}{500000000}$
a_{prune}	$\frac{36700631257}{50000000000}$
θ_1	$\frac{10059807341}{12500000000}$
ρ	$\frac{110002874129}{100000000000}$
w_A	$\frac{2311587725257500000000}{14597460907219352581}$
η	$\frac{244357}{2000000000}$
M	$\frac{10218421973}{10000000000}$
χ	$\frac{10077234539}{5000000000}$
B_0	$\frac{1225527173}{10000000000}$
`router_length`	$\frac{30589490361}{5000000000}$
`route_load`	$\frac{7519120361}{500000000}$
`order_bound`	$\frac{7065355693}{10000000000}$
C	$\frac{485}{42}$
σ	$\frac{23}{42}$
κ	$\frac{999999}{1000000}$
`edge_bias`	$\frac{511}{512}$
`split_deviation`	$\frac{1}{1024}$
p	$\frac{244357}{8000000000}$
`router_bank`	$\frac{1044483}{1048576}$
`completion_bank`	$\frac{1023}{1048576}$
`router_constant`	$\frac{52902787547485085406377454545012954383041}{625000000000000000000000000000000000000}$
ϕ_0	$\frac{46220729}{50000000}$

The archived ordinary-proof finite inputs are included for provenance. Their reported upper bounds can differ in the last few decimal places from the formally proved chain constants. This paper and its tables use the Lean constants exclusively; the index records the exact differences. No historical Python fee is used as a premise of the formal proof.

The main degree logarithm is 158387/1000. The tail parameter is $r = 88331750161/10^{11}$, with $\beta = r(1-r)$ and $Q = 1/\lfloor r^2(1-r)\rfloor$. The parameters `p`, `router_length`, `route_load`, `order_bound`, and `router_constant` belong to the completion and routing gates; they are not independent choices left open in the proof.

Rational certificates enclose powers of 2, logarithms, square roots and cube roots. For a positive series, a finite partial sum is a lower bound; an upper bound includes a proved geometric tail. The

renewal loss is bounded by a positive finite expansion with its proved tail, and all inequalities are preserved in the required direction. Tangent certificates control continuous density intervals; monotonicity and rate certificates control scale half-lines. These steps justify extending the finite data to all graphs and all numbers of stages.

The phase table below is for orientation: its displayed phase endpoints are rounded to six decimal places and are *not* used as certificate endpoints. The exact endpoints, thresholds $P_3, \dots, P_{16}$, and all $u_1, \dots, u_{16}$ are in `paper/data/phase-base-index.json` and in the corresponding Lean modules. The listed fees are the actual rational `costBound` constants of the Lean finite chains.

Phase	a (display)	d (display)	Chain	$f_{a,d}$
00	158.355466	159.607444	011	$\frac{5391841825851}{500000000000}$
01	159.607444	160.859421	069	$\frac{10796623618537}{1000000000000}$
02	160.859421	162.111399	012	$\frac{2702389777377}{250000000000}$
03	162.111399	163.363377	070	$\frac{2705625297969}{250000000000}$
04	163.363377	164.615354	014	$\frac{2708860390719}{250000000000}$
05	164.615354	165.867332	071	$\frac{10848385314567}{1000000000000}$
06	165.867332	167.119310	015	$\frac{10861328073341}{1000000000000}$
07	167.119310	168.371287	017	$\frac{2718571646969}{250000000000}$
08	168.371287	169.623265	043	$\frac{10887234417859}{1000000000000}$
09	169.623265	170.875243	000	$\frac{10900183002579}{1000000000000}$
10	170.875243	172.127220	073	$\frac{1364141621327}{125000000000}$
11	172.127220	173.379198	019	$\frac{2731521075901}{250000000000}$
12	173.379198	174.631176	059	$\frac{1367379994851}{125000000000}$
13	174.631176	175.883153	021	$\frac{10951994641041}{1000000000000}$
14	175.883153	177.135131	022	$\frac{10959542293243}{1000000000000}$
15	177.135131	178.387109	023	$\frac{10966965799873}{1000000000000}$
16	178.387109	179.639086	024	$\frac{2743615751629}{250000000000}$
17	179.639086	180.891064	044	$\frac{10982037016331}{1000000000000}$
18	180.891064	182.143042	026	$\frac{10989687427511}{1000000000000}$
19	182.143042	183.395019	076	$\frac{10997418068539}{1000000000000}$
20	183.395019	184.646997	051	$\frac{11005220797637}{1000000000000}$
21	184.646997	185.898975	052	$\frac{1376639403931}{125000000000}$
22	185.898975	187.150952	061	$\frac{1377635136691}{125000000000}$
23	187.150952	188.402930	032	$\frac{11029125826993}{1000000000000}$
24	188.402930	189.654908	077	$\frac{2759313584727}{250000000000}$
25	189.654908	190.906885	034	$\frac{11045464831563}{1000000000000}$
26	190.906885	192.158863	063	$\frac{138172005771}{12500000000}$
27	192.158863	193.410841	064	$\frac{11062149004901}{1000000000000}$
28	193.410841	194.662818	035	$\frac{11070615823097}{1000000000000}$
29	194.662818	195.914796	079	$\frac{2769792387169}{250000000000}$
30	195.914796	197.166774	038	$\frac{277194651861}{25000000000}$

Phase	a (display)	d (display)	Chain	$f_{a,d}$
31	197.166774	197.792762	039	$\frac{5546050006983}{500000000000}$
32	197.792762	198.105757	081	$\frac{11094248594251}{1000000000000}$
33	198.105757	198.262254	040	$\frac{11095327100827}{1000000000000}$
34	198.262254	198.340503	082	$\frac{1109587104771}{100000000000}$
35	198.340503	198.379627	067	$\frac{5548069169719}{500000000000}$
36	198.379627	198.399189	041	$\frac{11096276645863}{1000000000000}$
37	198.399189	198.408970	065	$\frac{2219267420263}{200000000000}$
38	198.408970	198.413861	047	$\frac{554818554329}{50000000000}$
39	198.413861	198.418751	066	$\frac{2774101376139}{250000000000}$

The 45 base intervals and their assigned chains are listed in the same exact index file. The following table records all 83 finite chains. Its cutoff x means the average-degree condition $2m \le 2^x n$; the file also records the larger integer degree cap used for this condition. The column u is the denominator of the harmonic starting bound, and s is the number of certified DFS steps. Every chain uses the same mixed-cover parameters $B_* = 43/10$, $A_* = 3$ and singleton weight $23/42$.

Chain	Exact cutoff x	s	u	Certified fee
000	$\frac{1091}{8}$	0	224502	$\frac{10900183002579}{1000000000000}$
001	$\frac{1129}{8}$	0	117478	$\frac{2191544772857}{200000000000}$
002	$\frac{118931}{1000}$	0	144130	$\frac{10674267180743}{1000000000000}$
003	$\frac{118933}{1000}$	0	144173	$\frac{2134858533271}{200000000000}$
004	$\frac{118941}{1000}$	0	144240	$\frac{10674395954517}{1000000000000}$
005	$\frac{11897}{100}$	0	144304	$\frac{2134954301487}{200000000000}$
006	$\frac{119239}{1000}$	0	145330	$\frac{5339126553537}{500000000000}$
007	$\frac{121397}{1000}$	0	153684	$\frac{10706185307421}{1000000000000}$
008	$\frac{12263}{100}$	0	157998	$\frac{2680536955283}{250000000000}$
009	$\frac{123863}{1000}$	0	163280	$\frac{2684527045207}{250000000000}$
010	$\frac{126329}{1000}$	0	173733	$\frac{1346254529237}{125000000000}$
011	$\frac{127383}{1000}$	0	178559	$\frac{5391841825851}{500000000000}$
012	$\frac{129381}{1000}$	0	187995	$\frac{2702389777377}{250000000000}$
013	$\frac{131261}{1000}$	0	197117	$\frac{10833912573677}{1000000000000}$
014	$\frac{131379}{1000}$	0	197460	$\frac{2708860390719}{250000000000}$
015	$\frac{133377}{1000}$	0	208051	$\frac{10861328073341}{1000000000000}$
016	$\frac{133727}{1000}$	0	209257	$\frac{2173173368277}{200000000000}$
017	$\frac{134377}{1000}$	0	213451	$\frac{2718571646969}{250000000000}$
018	$\frac{136193}{1000}$	0	223406	$\frac{5448912124509}{500000000000}$
019	$\frac{138373}{1000}$	0	236415	$\frac{2731521075901}{250000000000}$
020	$\frac{138659}{1000}$	0	237842	$\frac{10929793693067}{1000000000000}$

Chain	Exact cutoff x	s	u	Certified fee
021	$\frac{140371}{1000}$	0	248776	$\frac{10951994641041}{1000000000000}$
022	$\frac{141371}{1000}$	0	117483	$\frac{10959542293243}{1000000000000}$
023	$\frac{14237}{100}$	0	117477	$\frac{10966965799873}{1000000000000}$
024	$\frac{143369}{1000}$	0	117479	$\frac{2743615751629}{250000000000}$
025	$\frac{143591}{1000}$	0	117477	$\frac{1097613966829}{100000000000}$
026	$\frac{145367}{1000}$	0	117459	$\frac{10989687427511}{1000000000000}$
027	$\frac{146057}{1000}$	0	117478	$\frac{171797133157}{15625000000}$
028	$\frac{14729}{100}$	0	117476	$\frac{1375579270293}{125000000000}$
029	$\frac{148523}{1000}$	0	117477	$\frac{11014368472807}{1000000000000}$
030	$\frac{14867}{125}$	0	144185	$\frac{10674331494659}{1000000000000}$
031	$\frac{14876}{125}$	0	144442	$\frac{5337631723901}{500000000000}$
032	$\frac{150363}{1000}$	0	117493	$\frac{11029125826993}{1000000000000}$
033	$\frac{150989}{1000}$	0	117474	$\frac{11034209376807}{1000000000000}$
034	$\frac{152361}{1000}$	0	117445	$\frac{11045464831563}{1000000000000}$
035	$\frac{155359}{1000}$	0	117478	$\frac{11070615823097}{1000000000000}$
036	$\frac{155921}{1000}$	0	117479	$\frac{2768854402683}{250000000000}$
037	$\frac{15637}{125}$	0	168384	$\frac{10754072517109}{1000000000000}$
038	$\frac{157357}{1000}$	0	118851	$\frac{277194651861}{25000000000}$
039	$\frac{157857}{1000}$	0	119873	$\frac{5546050006983}{500000000000}$
040	$\frac{158231}{1000}$	0	120651	$\frac{11095327100827}{1000000000000}$
041	$\frac{158341}{1000}$	0	120834	$\frac{11096276645863}{1000000000000}$
042	$\frac{158387}{1000}$	0	120962	$\frac{11096672924573}{1000000000000}$
043	$\frac{16922}{125}$	0	218645	$\frac{10887234417859}{1000000000000}$
044	$\frac{18046}{125}$	0	117477	$\frac{10982037016331}{1000000000000}$
045	$\frac{18103}{125}$	0	117478	$\frac{2746379951309}{250000000000}$
046	$\frac{19336}{125}$	0	117477	$\frac{5532459781859}{500000000000}$
047	$\frac{19794}{125}$	0	120882	$\frac{554818554329}{50000000000}$
048	$\frac{2379}{20}$	0	144231	$\frac{10674512940043}{1000000000000}$
049	$\frac{23817}{200}$	0	144740	$\frac{1067625989843}{100000000000}$
050	$\frac{25759}{200}$	0	184710	$\frac{1350246585773}{125000000000}$
051	$\frac{29473}{200}$	0	117438	$\frac{11005220797637}{1000000000000}$
052	$\frac{29673}{200}$	0	117478	$\frac{1376639403931}{125000000000}$
053	$\frac{29733}{250}$	0	144141	$\frac{1067427985913}{100000000000}$
054	$\frac{29887}{250}$	0	146490	$\frac{10682252366379}{1000000000000}$
055	$\frac{30041}{250}$	0	148859	$\frac{5345112523633}{500000000000}$
056	$\frac{30691}{200}$	0	117472	$\frac{1105455351259}{100000000000}$
057	$\frac{32507}{250}$	0	190826	$\frac{10817943414113}{1000000000000}$
058	$\frac{3374}{25}$	0	216265	$\frac{5440921458393}{500000000000}$

Chain	Exact cutoff x	s	u	Certified fee
059	$\frac{34843}{250}$	0	242342	$\frac{1367379994851}{125000000000}$
060	$\frac{34973}{250}$	0	245655	$\frac{437831386747}{40000000000}$
061	$\frac{37341}{250}$	0	117473	$\frac{1377635136691}{125000000000}$
062	$\frac{37439}{250}$	0	117480	$\frac{11024226683983}{1000000000000}$
063	$\frac{3834}{25}$	0	117464	$\frac{138172005771}{12500000000}$
064	$\frac{3859}{25}$	0	117477	$\frac{11062149004901}{1000000000000}$
065	$\frac{39587}{250}$	0	120849	$\frac{2219267420263}{200000000000}$
066	$\frac{39589}{250}$	0	120885	$\frac{2774101376139}{250000000000}$
067	$\frac{6333}{40}$	0	120808	$\frac{5548069169719}{500000000000}$
068	$\frac{63781}{500}$	0	179618	$\frac{10786001135279}{1000000000000}$
069	$\frac{64191}{500}$	0	182773	$\frac{10796623618537}{1000000000000}$
070	$\frac{6519}{50}$	0	192519	$\frac{2705625297969}{250000000000}$
071	$\frac{66189}{500}$	0	202782	$\frac{10848385314567}{1000000000000}$
072	$\frac{66247}{500}$	0	203277	$\frac{5424943263449}{500000000000}$
073	$\frac{68687}{500}$	0	230378	$\frac{1364141621327}{125000000000}$
074	$\frac{68713}{500}$	0	230694	$\frac{109138064341}{10000000000}$
075	$\frac{71179}{500}$	0	117477	$\frac{10966874863597}{1000000000000}$
076	$\frac{73183}{500}$	0	117475	$\frac{10997418068539}{1000000000000}$
077	$\frac{75681}{500}$	0	117480	$\frac{2759313584727}{250000000000}$
078	$\frac{76111}{500}$	0	117476	$\frac{2761079337139}{250000000000}$
079	$\frac{78179}{500}$	0	117507	$\frac{2769792387169}{250000000000}$
080	$\frac{78577}{500}$	0	118427	$\frac{11086034684193}{1000000000000}$
081	$\frac{79053}{500}$	0	120382	$\frac{11094248594251}{1000000000000}$
082	$\frac{79147}{500}$	0	120696	$\frac{1109587104771}{100000000000}$

Appendix D. Correspondence with Lean

The following declarations are present in the audited transitive proof-body closure of the main theorem. The machine-readable table `docs/proof-map.json` records their modules and source paths. The conditional induction interface in the last part of the proof is instantiated with the proved physical construction; it is not an additional assumption of Theorem 1.1. Some source comments record earlier intermediate obligations. Their discharge is determined by the final theorem and its actual dependency chain, not by those historical comments.

Statement	Lean declaration
Final original-edge partition (1.1)	`Cycle12.coefficient12`
Rotation and deletion cycle bound (3.1)	`Cycle12.Independent.erdosGallai_cycle`
Bernoulli sample lower tail (3.2)	`Cycle13.vertex_sample_edge_lower_tail`

Statement	Lean declaration
Weighted order to coefficient twelve (2.3)	`Cycle12.coefficient12_from_weighted_order_bound`
Finite rooted amplification (2.2)	`Cycle12.rooted_weighted_order_to_predecessor`
Circumference at most six (5.1)	`Cycle12.terminal_rank_no_seven_cycle`
Harmonic affine extension (5.2)	`Cycle13.weightedAffine_harmonic`
Mixed cover (5.3)	`Cycle12.mixed_cover`
Certified DFS step (5.4)	`Cycle12.FiniteStageCertificate.step`
Finite-degree potential bound (5.5)	`Cycle12.Global12.finite_degree_potential`
Uniform loss decay (4.1)	`Cycle12.Global12.lossH_decay`
Renewal potential payment (4.2)	`Cycle12.Global12.globalPhi_fold_budget`
Actual even scale layers (6)	`Cycle12.Even12.exists_actual_scale_layers`
High-centre completion (7.1)	`Cycle12.Even12.high_center_completion_at_scale`
Even-bank grouped weighted routing (7.2)	`Cycle12.Even12.bank_routing_at_scale`
Joint original endpoint family (7)	`Cycle12.Even12.ActualFoldStage12.jointFamily`
Active mass lower bound (35)	`Cycle12.Even12.ActualFoldStage12.activeMass_lower`
Four-family actual packing payment (8.1)	`Cycle12.Even12.ActualFoldStage12.actual_fold_target_payment`
Whole-graph physical construction (8.1)	`Cycle12.Global12.physical_fold_construction12`
Physical scalar ledger (8.2)	`Cycle12.Global12.physical_fold_ledger_payment`
No-fold core mass bound (9.1)	`Cycle12.Even12.EvenScaleLayers12.kept_mass_of_no_fold`
Infinite terminal tail (9.2)	`Cycle12.Global12.infinite_terminal_tail_payment`
Terminal potential payment (9.2)	`Cycle12.Global12.terminal_potential_payment`
Strong induction from physical branch (10.1)	`Cycle12.Global12.potential_bound_from_fold_branch`
Coverage of finite parameter tuples (A)	`Cycle12.six_residue_parameter_coverage`
Full continuous cover (B)	`Cycle12.CoverAssembly.Full.sound`

References

1. Ron Aharoni and Penny Haxell, *Hall's theorem for hypergraphs*, Journal of Graph Theory **35** (2000), no. 2, 83–88.
2. Matija Bucić and Richard Montgomery, *Towards the Erdős–Gallai cycle decomposition conjecture*, Advances in Mathematics **437** (2024), 109434, arXiv:2211.07689.

3. Ryan Coffey, *A proof of the Erdős–Gallai cycle decomposition conjecture*, https://github.com/steelwheel01/erdos-gallai-lean, 2026, Manuscript and Lean formalization; repository consulted 3 November 2026.
4. David Conlon, Jacob Fox, and Benny Sudakov, *Cycle packing*, Random Structures & Algorithms **45** (2014), no. 4, 608–626.
5. Leonardo de Moura and Sebastian Ullrich, *The Lean 4 theorem prover and programming language*, Automated Deduction—CADE 28 (André Platzer and Geoff Sutcliffe, eds.), Springer, 2021, pp. 625–635.
6. Paul Erdős and Tibor Gallai, *On maximal paths and circuits of graphs*, Acta Mathematica Academiae Scientiarum Hungaricae **10** (1959), 337–356.
7. Paul Erdős, A. W. Goodman, and Louis Pósa, *The representation of a graph by set intersections*, Canadian Journal of Mathematics **18** (1966), 106–112.
8. C. M. Fortuin, P. W. Kasteleyn, and J. Ginibre, *Correlation inequalities on some partially ordered sets*, Communications in Mathematical Physics **22** (1971), no. 2, 89–103.
9. G. Gauthier, T.-N. Le, and P. Wollan, *Forcing clique immersions through chromatic number*, European Journal of Combinatorics **81** (2019), 98–118, arXiv:1703.08235.
10. David S. Gunderson, *Graph theory notes*, https://mspace.lib.umanitoba.ca/bitstream/handle/1993/37010/GraphTheoryNotes2022.pdf, 2022, University of Manitoba, 8 December 2022, Theorem 2.7.1, pp. 104–105.
11. Ervin Győri, Tamás Róbert Mezei, and Gábor Mészáros, *Terminal-pairability in complete graphs*, https://arxiv.org/abs/1605.05857, 2018, Journal of Combinatorial Mathematics and Combinatorial Computing 107; arXiv:1605.05857v2.
12. Jaehoon Kim, *A proof of the Erdős–Gallai cycle decomposition conjecture*, https://arxiv.org/abs/2610.07840, 2026, arXiv:2610.07840v2, 9 October 2026.
13. László Lovász, *On covering of graphs*, Theory of Graphs (Proceedings of the Colloquium, Tihany, 1966) (Paul Erdős and G. Katona, eds.), Academic Press; Akadémiai Kiadó, 1968, pp. 231–236.
14. OpenAI, *A linear cycle-and-edge decomposition of every graph*, https://github.com/openai/math/tree/main/preprints/A-linear-cycle-and-edge-decomposition-of-every-graph-September-24-2026, 2026, Preprint, 24 September 2026. OpenAI Math Release.
15. The mathlib Community, *The Lean mathematical library*, Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs, CPP 2020, ACM, 2020.
16. Douglas R. Woodall, *Sufficient conditions for circuits in graphs*, Proceedings of the London Mathematical Society **24** (1972), no. 4, 739–755.

Acknowledgements and use of AI tools

OpenAI ChatGPT/Codex tools were used extensively in the development of the argument, computational exploration, preparation of the Lean formalization, and drafting and organization of this manuscript. The source repository records the formal theorem, its proof dependencies, and the scope of the reported checks. These tools are not authors; responsibility for the submitted work rests with the author. The author also acknowledges the contributors to Lean, Mathlib, and the third-party formal libraries identified in the repository.